

FIT2107 Unit Primer

Testing is a critical part of software development. Through this unit you will learn how to define what quality means for a piece of software, as well as common testing techniques and how to implement them in code.

Unit learning outcomes

To complete this unit, you need to demonstrate that you can do the following:

- ULO 1: Identify and prioritise quality attributes for a software system;
- ULO 2: Design test cases using black box and white box techniques;
- ULO 3: Select appropriate quality assurance strategies and testing techniques to develop and document software testing plans;
- ULO 4: Implement software tests following a documented plan, using modern programming techniques and best practices;
- ULO 5: Automate the execution of software tests and code analysis using continuous integration frameworks;
- ULO 6: Report code quality issues using contemporary, data-driven code review practices.

Through the unit you will be working on a portfolio of tasks that shows you have these capabilities. At the end of the unit you will reflect on your achievement of these Unit Learning Outcomes (ULOs) when you finalise your portfolio for submission.

Meaning of grades

Different grades require you to demonstrate different skills. In this unit there are only two options: pass and fail.

1. **Pass:** has a foundational understanding of software quality and testing.

Through the **Pass Tasks** you will develop an understanding of basic testing techniques and how to apply them to simple problems. You will also demonstrate an understanding of how the concepts you have learned relate to each other.

2. **Fail:** has not demonstrated the required competency across all learning outcomes.

Failing to submit a portfolio that satisfactorily meets all of the hurdle requirements (including performance on pass tasks and task discussions/interview) will result in a fail grade being awarded. To avoid this, make sure you submit tasks regularly, work ahead of the task deadlines, respond to feedback, and attend class regularly to discuss your progress. There is plenty of feedback and support to help you pass – the main thing is to engage actively with your learning and the teaching team throughout the semester.

Feedback

Feedback provides you with information on your work, and guidance on how to improve. Submitting your work in OnTrack will start the feedback process. **Your tutor will aim to look at any tasks you submit before their feedback deadline within a week of submission.** They will review your work and respond with comments. They will then change the task status to indicate what you need to do next. For example, you may need to fix and resubmit aspects of your work, or come into class for a discussion.

You need all tasks marked as Complete to guarantee passing the unit. This requires you to:

- **submit** the task ahead of its feedback deadline,
- **respond** to any feedback and **resubmit** the work if required,
- attend class to **discuss** the task and how you went with it.

You might need to do this more than once: most students need to submit a task 2 – 3 times before it is ready for discussion. So, we encourage you to **submit tasks early** so you have enough time to resubmit (possibly more than once!) and still get more feedback.

We're here to help

Think of the teaching team as your personal learning coaches. We will push you to achieve the most you can.

Work submitted after a task's final feedback date will automatically be marked as **Assess in Portfolio**. This means you won't receive feedback in OnTrack, and you won't have an opportunity to discuss the task with staff and get it marked as Complete. While this does happen sometimes, it should only be due to truly exceptional circumstances (in which case consider applying for special consideration!). Plan to get your work submitted well ahead of the feedback deadlines to ensure you have the time to discuss the task, address feedback, and resubmit where needed.

Task discussions

It is important that you are able to demonstrate your capabilities. To achieve this you **must** discuss tasks in your applied class in order to get them signed off as complete.

Your task will be marked as **Discuss** in OnTrack if you submit it before the final feedback date and your submission indicates that you appear to have achieved the required competency. Once this happens, you must attend your applied class and discuss this task with your TA **within two weeks**, otherwise the task will be moved to **Fix and Resubmit**.

Your job in these task discussions is to convince the teaching team that you have developed the competencies required by the task. This should not be difficult if you have engaged with the learning resources and tackled the task yourself, and the process will also help enhance your learning if there are some aspects that you have missed or misunderstood. Where teaching staff are not clearly convinced of your competency they may ask you to do additional work, resubmit the task, and/or discuss the task again. Work with the teaching team and you will develop the level of competency you are aiming to achieve.

Submission limits

Resubmissions are an important part of learning in this unit, as they give you the opportunity to demonstrate that you have learned from your misunderstandings or mistakes. In general you will be able to resubmit each task as many times as you need. However, if you abuse this opportunity and are not demonstrating sufficient progress between resubmissions, staff have the option to set your task to **Assess in Portfolio** and exclude you from receiving any further feedback.

Minimum pass criteria

To pass this unit you must submit a portfolio that contains:

- All pass tasks to a standard that demonstrates you have the required competencies.
- Sufficient tasks marked as Complete to evidence task discussion and validate the competency shown in your submitted work.

Where there is insufficient evidence of the above to convince the portfolio assessment panel that you have the required competencies, you will not be eligible to pass the unit.

If your work is close to demonstrating appropriate competency, you may be invited to attend supplementary assessment. Please note that sitting supplementary assessment will limit your final result to 50P, as indicated in the [supplementary assessment policy](#).

Timelines

The OnTrack task planner will show you a suggested plan for completing the unit tasks. This is the slowest possible path, and leaves no buffer. We strongly suggest that you adjust this plan to aim to finish tasks earlier, and to align with your personal circumstances and commitments. Then, regularly revisit this page to see if you are on track with your plan and update it if needed.

Learning with others, and generative artificial intelligence

Learning with others in this unit is not only promoted but expected. Learning with peers through asking for help, supporting others, or sharing your learning has many benefits for all involved, such as social support and encouragement. However, you need to be careful to **avoid collusion**!

What is collusion? Copying or letting someone copy your work. For that reason, when you give or receive help, make sure the conversation focuses on explanations of how things work, how to do things, or how to think about addressing the task, not on completed solutions.

Similar advice applies for collaborating with generative artificial intelligence (GenAI). You may choose to engage with GenAI to help your learning, but **avoid asking it to do the work for you**. Just like you won't learn much if we give you the answers, you won't if GenAI gives you them either. You need to develop your own skills. If you overly rely upon GenAI you may not be able to adequately discuss your work. This will place you in significant risk of failing the unit. There are many supports in place that can help you achieve the required capabilities, so engage with these resources when you become stuck.

How to succeed in FIT2107 Software Quality and Testing

Attend classes: The applied classes give you access to teaching staff and other students working on the same tasks. This is a great environment to engage with the unit, clarify difficulties, and accelerate your progress.

Students who attend class regularly are much more likely to find the OnTrack tasks easier to complete, stay up to date, and generally find the unit manageable. Students who miss classes tend to take much longer to complete OnTrack tasks find them more difficult to complete, and are much more likely to fail the unit.

Stay up to date with your tasks: plan out how you will complete the unit tasks and work well ahead of the feedback deadlines. Leaving the tasks to the last minute runs the risk of missing feedback, and ending up with a portfolio where insufficient tasks are marked as Complete.

Engage and be involved: successful students are those who are actively engaged. Ask questions, be curious, be active in class and online, and reach out for help when you need it. You'll learn a lot from these interactions, so make yourself known to the teaching team and your classmates.